Looped Transformers and Recurrent Depth

Eight years from Universal Transformers to GPT-6 Astra (2018–2026): five core papers and ten trend insights

Presenter: awesome_loop_transformer project

github.com/asimfish/awesome_loop_transformer

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Outline

1. Why now
2. Five core papers
3. The wider map and ten insights
4. Open problems and takeaways

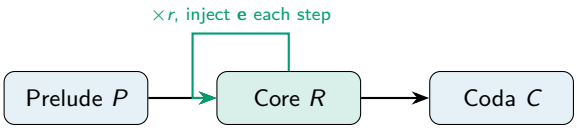
A 3.5B model that spends FLOPs like a 32B one

Reported, September 2026: The Information says GPT-6 Astra uses "recurrent depth".

The idea

Apply **one set of parameters** repeatedly along depth:

$$\mathbf{s}_i = R(\mathbf{e}, \mathbf{s}_{i-1}), \quad i = 1, \ldots, r$$



Source: Geiping et al. 2025, Fig. 2 (redrawn)

- Depth $2 + 4r + 2$ grows with r; parameters stay at 3.5B
- r=4 → 32: ARC-E 49.1 → 69.9, GSM8K-CoT ≈ 0 → 42.1 (Huginn)
- No chain-of-thought data; small context window

Takeaway: recurrence trades parameters for compute; this talk asks when that trade wins.

Taxonomy: where the loop sits, and three orthogonal axes

Granularity	Definition	Representatives
model-loop	whole stack fed back	UT (2018), Huginn core, Ouro
block-loop	contiguous middle segment loops	Huginn 3-stage, Hyperloop, Training-Free, SMELT
layer-loop	each layer repeats K times, then next layer	Loopie, Training-Free layer-mode, MixerLoop
latent conditioning	previous state injected as condition	Coconut, PLT (CLP), Latent Recurrent Tr.

Loop count
fixed (Loopie, LoopCoder-v2) / random-train, free-test (Huginn) / per-token halting (ACT, MoR, TaH)

Training
from scratch (Huginn, Ouro, Loopie) / retrofit (Relaxed Recursive, Retrofitted Recurrence) / training-free

What the loop is
unrolled LM / fixed point (DEQ, STARS) / ODE refinement (Training-Free) / supervised latent CoT (LOTUS) / RL on steps (RLTT)

Universal Transformers (2018): recurrence + adaptive halting

Mechanism (Dehghani et al., ICLR 2019)

$$A^t = \mathrm{LN}\big(H^{t-1} + P^t + \mathrm{MHSA}(H^{t-1} + P^t)\big)$$
$$H^t = \mathrm{LN}\big(A^t + \mathrm{Transition}(A^t)\big)$$

- Parameters shared over time steps t; coordinate embedding P^t = position + time step
- ACT halting per position: ponder time 2.3 / 3.1 / 3.8 steps for 1 / 2 / 3 supporting facts
- Turing-complete given memory (reduces to Neural GPU)

Task	Trans.	UT	+ACT
bAbI 10K joint (err%)	22.1	0.47	**0.29**
LAMBADA ppl	7321	319	**142**
WMT14 En-De BLEU	28.0	**28.9**	–

Open problems it left: parameter–compute ratio, training stability, inference cost.

Takeaway: gains concentrate on hard instances; the same pattern reappears in Huginn seven years later.

Huginn-3.5B (Feb 2025): recurrent depth trained at modern scale

- Shape $(2, 4, 2)$, $h = 5280$, 3.5B params; $r = 32 \Rightarrow 132$ effective layers
- Train: $r \sim$ log-normal Poisson ($\bar{r} = 32$), backprop through last $k = 8$ steps only
- 800B tokens, 4096 AMD MI250X, constant LR 4×10^{-5}, sandwich norm
- Two failed runs: hidden-state collapse; recurrence collapse (r=1 ≡ r=32)

Fair comparison is at 0.18T tokens (Table 4)

Same setup, non-recurrent: ARC-E 46.42, GSM8K-CoT 2.20 vs. recurrent r=32: **53.62**, **10.24**

r	ARC-E	HellaSwag	MMLU	GSM8K CoT
4	49.07	43.46	23.39	–
8	65.11	58.54	25.29	–
16	69.49	64.67	31.25	–
32	**69.91**	**65.21**	**31.38**	**42.08**

Table 1–2; GSM8K CoT strict, with system prompt.

The last 620B tokens improve r=32 but not r=1 (34.01 → 34.89): everything learned lives in the loop.

Huginn: what the loop buys, and what emerges in latent space

Zero-shot capabilities

- Per-token adaptive exit ($\mathrm{KL} < 5{\times}10^{-4}$): MT-Bench 5.63 $\rightarrow$ 5.56
- KV-cache sharing with budget 4: no loss
- Continuous CoT: warm-start $\mathbf{s}_0$ from previous token
- Self-speculative decoding without a draft model

- More context $\Rightarrow$ more loops needed: ARC-C saturates at 8 (0-shot) vs. 32 (25–50-shot)
- Open-book OBQA +11.0: reasons over context, memorizes little

Latent geometry (Sec. 7)

Fixed points for most tokens; **orbits** for numeric tokens; **sliders** (steady drift) for key tokens; path independence across $\mathbf{s}_0$.

Caveats: one run, LR never cooled; $r{=}32$ inference is $16\times$ an 8-layer model; Lu et al. 2025 find limited interpretable latent CoT.

Takeaway: "3.5B behaves like 50B" is FLOP-equivalence, not benchmark-equivalence.

Training-Free Looped Transformers (May 2026): a layer is one Euler step

Insight (Chen, Li, Liang, Lao, Liu; UT Austin)

Pre-norm layer $L(x) = x + F(x)$ is forward Euler (h=1) on $\dot{x} = F(x)$. Naive K-fold looping integrates to t=K — post-loop layers never saw that.
Sub-step instead:

$$x_{k+1} = \left(1 - \tfrac{1}{K}\right) x_k + \tfrac{1}{K}\, g(x_k)$$

- RK tableau with anchor β: output $= \beta\, g(x_0) + (1-\beta)\, F^K(x_0)$
- Toy net, K=8: naive MSE 335 vs. sub-step 1.04; accelerators (Anderson, heavy-ball, Aitken) never beat RK $\Rightarrow$ mid-stack is an ODE flow, not a contraction

Model	Benchmark	Base	Δpp
Qwen3-4B-Instruct	MMLU-Pro	57.1	**+2.64**
Qwen3-4B-Instruct	GPQA-Main	33.7	**+2.01**
Qwen1.5-MoE-A2.7B	ARC-C	48.3	**+2.30**
Llama-3.2-1B-Inst.	GPQA-Main	27.9	+1.79
Moonlight-16B-A3B	OBQA	31.6	+1.20
DeepSeek-V2-Lite	ARC-C	57.9	+0.85

One recipe, no per-cell tuning: middle 4 layers, 3-stage RK, block-mode dense / layer-mode MoE.

Training-Free: window size, depth fraction, and the MoE rule

Window n	Layers	Δ (16-task)
1	[14]	+0.18
3	[13–15]	+0.21
4	[12–15]	**+0.55**
6	[11–16]	−0.82
28	all	−27.73

Qwen3-1.7B-Base, forward Euler K=2 (Table 2a).

- **Depth-fraction rule**: optimal window center at 0.43–0.71 of depth (>1.7B), 0.25–0.56 (<1B)
- **MoE needs layer-mode**: block-mode re-routes experts each iteration (routing thrash); layer-mode gates once
- 45 cells: 60% positive, 27% neutral, 87% non-negative; failures = sub-3B distilled on knowledge MC
- Naive loop at K=6: −17.71pp; this method stable for $K \leq 24$

Takeaway: an untrained model can only be looped in a small mid-stack window; gains cap at ∼2–3 points.

LoopCoder-v2 (Jun 2026): in a Parallel Loop Transformer, loop 2 is almost everything

PLT (Wu et al. 2025) flattens cost

- Shared first-loop KV + gated sliding window (w=64): KV $O(LSd)$
- Cross-loop position offset: $B^{(r)} = \mathrm{Embed}(x) + \mathrm{shift}(\mathbf{h}^{(r-1)}) \Rightarrow$ loops run in one pass
- Price: token i receives its neighbor's state (positional mismatch)

7B	HE+	LCB	SWE-V	Multi-SWE	Avg
R=1	81.1	27.4	43.0	14.0	38.0
R=2	**84.1**	**35.4**	**64.4**	**31.0**	**46.5**
R=3	75.0	28.6	27.6	11.0	36.9
R=4	76.8	24.5	22.4	9.3	34.3

Table 2. R=2 on SWE-bench Verified (64.4) beats Kimi-Dev-72B (60.4) and Qwen3-235B-A22B (45.2).

7B dense, $R \in \{1, 2, 3, 4\}$ each trained from scratch on 18T tokens; same SFT; 1M GPU hours.

Takeaway: strongly non-monotonic in R — the next slide shows why.

LoopCoder-v2: gain–cost scissors, and CoT + loop is super-additive

Loop	$\delta^{(r)}$	$\Delta p^{(r)}$	eff. rank	$\cos\theta$
r=2	**846**	**1.75**	**174.6**	−0.72
r=3	464	1.32	172.5	−0.46
r=4	1014	1.58	158.2	0.04

PLT4, 500 samples (Table 3). Offset cost $\Omega^{(r)} = \frac{1}{S}\sum_i \|\mathbf{h}_i^{(r-1)} - \mathbf{h}_{i-1}^{(r-1)}\|$ is $\approx$ constant.

- Effective rank peaks at loop 2, then narrows
- Updates oscillate ($\cos\theta < 0$); attention routing freezes after loop 2
- Fixed offset tax is **30–45**$\times$ the marginal gain from loop 3 on

Thinking mode on top of R=2 (Table 4)

	Instruct	Thinking
LiveCodeBench	35.4	**62.3**
CRUX	86.9	**93.5**
MultiPL-E	73.9	**77.8**

Explicit CoT alone does not help the non-looped model; loop alone gives single digits. Together: super-additive.

Scope: "two is optimal" is a PLT property (CLP tax). Huginn improves monotonically to r=32.

Loop the Loopies! (Jul 2026): the first compute-matched win

Loopie Recipe (match by measured step time)

1. Halve stored layers, loop twice: LR unchanged, activation memory $\propto sbDL$ halves
2. Spend memory on $2\times$ micro-batch (tokens/step unchanged)
3. Reinvest measured speed-up into width; pick equal wall-clock step time

Config	D	L	R	$\hat{C}$	$\hat{M}_{\text{act}}$
Qwen3-30B-A3B ref.	2048	48	1	1.00	1.00
seed Loopie	2048	24	2	1.00	0.50
Loopie-20B-A2B	2304	27	2	**1.42**	0.63

- **Layer-loop** ($L_1 L_1 L_2 L_2 \dots$) overtakes model-loop after $\sim$1.2T tokens; pipeline-parallel friendly
- Compute-matched run: overtakes reproduced Qwen3-30B-A3B after $\sim$600B of 800B tokens
- Scaling ladder 0.15B–1B active params: Loopie wins all 4 rungs, gap does not shrink
- Why R=2: marginal return of $N\times$ loops vs. $N\times$ layers decays fastest after 2

Accounting: nominal FLOPs +42%; inference does 54 vs. 48 block passes per token; no latency reported.

Loopie: post-training with supervised pre-training (SPT) and RL

SPT = SFT loss mask + pre-training scale

Loss on target tokens only; global batch $\geq$1024, sequence $\geq$128K, $\geq$128M token positions per batch, 2T tokens, 10 epochs. Pretraining metrics $\uparrow$ and reasoning metrics $\uparrow$; no overfitting, no forgetting.

RL: GSPO (sequence-level ratio/clip) + DAPO clip-higher and dynamic sampling; math then code; 32K $\rightarrow$ 64K.

	Qwen3 30B-A3B (36T)	Nemotron 3 Nano (25T)	**Loopie 20B-A2B (3.5T)**
MMLU	85.83	80.52	**81.28**
ARC-C	96.67	91.96	93.52
BBH	86.59	68.76	82.28
IFEval	85.58	77.05	84.72
AIME 24	90.10	85.00	**92.09**
AMC	96.73	91.80	94.21

Table 3. Loopie-6B-A0.6B Thinking: AIME24 80.42, AIME25 70.83 (Ouro-2.6B: 62.50 / 51.67).

Takeaway: under one-seventh of the pretraining tokens, a looped MoE matches 25T-token peers on knowledge benchmarks.

The 2026 literature: ten themes, roughly sixty papers

Theme	Representative results
Theory	k layers looped L times $\approx kL$ layers (Saunshi 25); loops vs. CoT separation (Xu & Sato 25); budget law $v \approx n/T$ (Zhang 26)
Scaled pretraining	Huginn 3.5B/0.8T, Ouro 2.6B/7.7T, LoopCoder-v2 7B/18T, Loopie 20B-A2B/3.5T, Nanbeige4.2-3B/28T, SMELT to 54B; loop counts 2–4
Efficiency	PLT (latency), diffusion-style sampler 5×, MELT constant KV, Looped Latent Attention 32× KV, LT2 linear attention, LoopQ
Stability	residual scaling $\varepsilon{=}1/N$ (not $1/\sqrt{L}$), DeepLoop exponent $1/4 \to 1/2$, STARS spectral radius, settling/drifting regimes
Adaptive halting	MoR (NeurIPS 25), Think-at-Hard (skip 93% of tokens), LoopFormer, RecurTrace (2.0 loops avg.)
Retrofit / training-free	Relaxed Recursive (LoRA), Retrofitted Recurrence, Training-Free RK, Program-of-Layers (ICML 26)
Post-training	RLTT trajectory credit +10.9%, LoopRPT, LOTUS latent CoT matches explicit at 3B, SPT
Interpretability	orbits/sliders (Huginn), limited latent-CoT evidence (Lu 25), Jacobian lens: Ouro rebuilds workspace per loop
Tiny reasoners	HRM 27M, TRM 7M (ARC-AGI-1 45%), Attractor Models, Denoising Recursion
Applications	LoopViT (ARC 65.8% at 18M), RD-VLA (0% → 90%, 80× faster), compositional tool calling, agentic products

Insights 1–3: narrow win, two regimes, converging on the middle

1. **Compute-matched wins exist, through a narrow door.** Loopie (step time) and SMELT (FLOPs/params/KV, 6.8–18% savings) win only with MoE + layer/mid-stack loop + two loops. One recurrence is worth $r^{0.46}$ unique parameters (0.38 under truncated backprop): loops cannot replace parameters; Loopie converts *activation memory* into width.
2. **"How many loops" splits by where compute binds.** Fixed pretraining budget $\Rightarrow R{=}2$ (Loopie, LoopCoder-v2, SMELT, Nanbeige). Variable inference compute $\Rightarrow$ adaptive and deep (Huginn: 8 loops zero-shot, 32 with 25–50 shots). "Three regress" is PLT's positional tax.
3. **Granularity shrinks, placement converges.** Whole model $\rightarrow$ three-stage $\rightarrow$ per-layer / mid-stack $\rightarrow$ mixer-only (MixerLoop: 41.5% of gain at -45.9% looped FLOPs). Optimal window center ≈ 0.5 of depth; MoE needs layer-mode (two independent papers agree).

Insights 4–7: ODE vs. fixed point, stability theorems, inference cost, CoT complementarity

4. **Two philosophies.** Training-Free: a pretrained mid-stack is an ODE flow (sub-step, keep the endpoint). STARS / Fixed-Point Reasoners / Attractor Models: *train* the loop to be a fixed point. Viakhirev et al.: settling operators are depth-safe; Huginn is non-settling; LoopCoder-v2's $\cos\theta < 0$ is drift.
5. **Stability went from folklore to theorems.** $\varepsilon{=}1/N$ residual scaling; DeepNorm exponent $1/2$; LR depends only on unique layers and transfers across N; stable training to 12–15 loops.
6. **Inference cost decomposes into three stackable fixes.** Latency (PLT, diffusion sampler $5\times$), KV (sharing; LLA low-rank $32\times$), quantization (LoopQ). Matched training compute $\neq$ matched inference cost.
7. **Loops and CoT are complements.** LoopCoder-v2 LCB $+26.9$ super-additive; LOTUS matches explicit CoT at 3B with 2.5–6.9$\times$ lower thought latency. Standard GRPO credits only the final latent state: RLTT ($+10.9\%$) and LoopRPT redesign credit assignment.

Insights 8–10: capability profile, applications, productization

8. **Loops manipulate knowledge; they do not store it.** Ouro attributes gains to knowledge manipulation; Huginn open-book +11.0; Frey et al.: loops help math, memory banks help commonsense; Dual-Path gates send arithmetic tokens "deep" and function words "wide". Best partners: retrieval, long context, MoE expert banks.
9. **From puzzles to agents and embodiment.** SWE-bench Verified +21.4 (LoopCoder-v2); compositional tool calling improves with depth; Nanbeige4.2-3B beats Qwen3.5-9B on agentic benchmarks; RD-VLA 0% $\rightarrow$ 90% with four iterations, 80$\times$ faster than token-reasoning VLAs.
10. **Productization started; tooling lags.** Five open families (Huginn, Ouro, LoopCoder-v2, Loopie, Nanbeige4.2-3B). Nanbeige4.2-3B ships with five out-of-the-box bugs in HF transformers and 2$\times$ attention memory from layer reuse. No standard interface yet for looped inference, KV, or quantization.

Five circulation errors, checked against the papers

Common claim	What the paper says
Huginn crushes its baseline "at the same 0.8T tokens"	Baseline trained 0.18T; at 0.18T recurrent ARC-E 53.62 vs. 46.42 (Table 4)
"Two loops suffice, three regress" is a law	PLT-specific (CLP tax), each R trained separately; Huginn monotone to r=32
Loopie and Qwen3-30B-A3B use "the same compute"	Same measured step time; nominal FLOPs +42%; inference cost unreported; reference is an 800B-token reproduction
Training-Free makes open models "stronger"	+1–2.6pp on knowledge MC; sub-3B distilled models can lose; +15–30% latency
Huginn "performs like 50B at 3.5B"	FLOP-equivalent; on benchmarks near OLMo-7B, behind OLMo-2-7B

GPT-6 Astra: what is verifiable and what is inference

Not verifiable here

Which loop Astra uses, how many, how trained. OpenAI's page discloses no architecture; The Information's report is anonymously sourced; neither was accessible to us.

- **Inference 1.** The public recipe closest to "frontier scale + compute-matched" is Loopie/SMELT: MoE, mid-stack or per-layer, small loop count.
- **Inference 2.** "OpenAI beat two-loop saturation" starts from a false premise — saturation is PLT-specific; the real bottleneck is inference cost, already cut $\sim 10\times$ by public KV compression and parallel sampling.
- **Inference 3.** Reported "security concerns" match a real gap: latent reasoning has no readable intermediate tokens; probes depend on layer and decoding method.
- **Inference 4.** Loopie (Jul), LLA (Jul), SMELT (1 Sep), Astra (early Sep): several groups reached usability simultaneously.

Open problems

1. Compute-matched evidence for $R \geq 4$ (all wins are at R=2)
2. The full inference bill: step time + latency + peak KV in one paper
3. Scaling-law form at 100B+ (SMELT to 54B, Schwethelm to $\sim$1B)
4. Adaptive halting beyond 8B (only Ouro's entropy allocation at 7.7T)
5. Supervisability of latent reasoning without gold CoT (safety)
6. Mechanism of layer-loop (only SMELT's attention-sink account)
7. Loops vs. sequence-length scaling (Hidden Decoding, WeLM 617B) at 100B+

Most valuable gaps

Compute-matched scaling laws at $R \geq 4$; loop-specific RL objectives with trajectory-level credit.

Three takeaways

1. **Recurrent depth passed the compute-matching gate in 2026** — via layer-loop, MoE, and two loops, with the win coming from activation memory, not from loops replacing parameters.
2. **Pick the loop count by where compute binds**: 2 for frontier pretraining; adaptive and deep when inference compute is variable. Two-loop saturation is PLT's price, not a ceiling.
3. **Loop under CoT, not instead of it**; redesign RL credit for latent trajectories; measure latency and KV before shipping.

Practical recipe

Zero-training: middle 4 layers, 3-stage RK, β mid, layer-mode on MoE. From scratch: MoE, layer-loop, R=2, match by measured step time, $1/N$ residual scaling, SPT + trajectory RL. Variable inference: sequential loop, random-r training, truncated backprop, adaptive exit, KV codec.

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Thank you

Full reports (ZH/EN), translated PDFs, HTML deck and paper list:
`https://github.com/asimfish/awesome_loop_transformer`

Backup Slides

Backup: Huginn training details

- Data: 800B tokens (795B realized), public, skewed to code/math; instruction data mixed into pretraining; vocab 65536; sequence 4096
- Hardware: ORNL Frontier, 4096 MI250X, 21 segments $\leq$12h; 108.75 TFLOP/s per node (87% achievable), 52–64 TFLOP/s per GPU at scale; data parallel + optimizer sharding only
- Optimizer: Adam (0.9, 0.95), constant LR 4×10^{-5} after 4096 warm-up steps, no decay; locked-step sampling of r per micro-batch
- Init: Takase et al. 2024, variance $2/(5h)$; output projections $1/(5hl)$ with $l = 132$
- Norm: four RMSNorms per layer ("sandwich"); RoPE base 50000; gated SiLU MLP; QK biases
- EMA over last 75 checkpoints: GSM8K flexible 47.23% at r=64

Backup: Training-Free RK tableau and cost

$$h = 1,\ s = K,\ a_{ij} = \tfrac{1}{K}\,\mathbb{I}(j < i), \qquad b_1 = \beta + \tfrac{1-\beta}{K},\ b_i = \tfrac{1-\beta}{K}\ (i \geq 2)$$

$$x_1 = x_0 + \sum_{i=1}^{K} b_i k_i = \beta\, g(x_0) + (1-\beta)\, F^K(x_0), \qquad F(x) = x + \tfrac{1}{K} F_g(x)$$

- Extra cost: $(b - a + 1)(K - 1)$ layer forwards; K=3 on a 4-layer window $\approx$ 8 extra layers (15–30% latency on 28–48-layer models)
- Seven strategies compared: naive, forward Euler, midpoint/Heun, RK4, heavy-ball, Anderson, Aitken, Uniform Loop
- Leakage-free transfer to Qwen3-30B-A3B at window $[22, 24]$: 8 of 9 benchmarks improve (CSQA $+1.14$, MMLU-flex $+0.79$)

Backup: Loopie pretraining and RL details

- Stage 1: Nemotron-CC-v2-HQ, $\sim$570B unique tokens, 4 epochs (2.28T); Stage 2: 1.26T annealing (SFT-v1 27.8%, Specialized-v1 21.9%, Code-v2 20.7%, HQ-Synthetic 15.6%, Math-v1 10.0%)
- Qwen3 tokenizer; AdamW ϵ=10^{-15}, wd 0.1, LR 5×10^{-4} (6B) / 3×10^{-4} (20B), warm-up-stable, batch 1024×8192
- Init: embeddings $\mathcal{N}(0, d^{-1})$; block parameters SmallInit $\mathcal{N}(0, 2/(5d))$
- RL: GSPO sequence-level ratio $s_i(\theta) = (\pi_\theta / \pi_{\theta_{\text{old}}})^{1/|o_i|}$ with asymmetric clipping; prompt-level dynamic filtering; Guru-RL math and code pools; 32K then 64K response length
- Code released as megatron-loopie and vllm-loopie forks